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Analytical Cume

March 31, 2006

There are three basic methods for determining an unknown concentration using a set of standards, calibration curve, standard addition and internal standard. Answer the questions below based on your basic knowledge of each procedure.

I. Calibration Curve

Absorbance at 1250 cm^{-1} for acetone in solutions in water are shown below:

Volume % acetone	Corrected Absorbance (A)
10	0.215
20	0.411
30	0.578
40	0.745
50	0.913

The resulting equation of the calibration curve is given by:

$$A = 0.01730\ (\pm 0.00033)\ \text{Vol\%} + 0.053 (\pm 0.0104)$$

(12 pts) a) Find the volume percent of acetone in a solution whose absorbance is 0.611

(12 pts) b) If the uncertainty in the measured absorbance is ±0.0104, what is the uncertainty in the calculated percentage of acetone, rounded off to the correct number of significant figures?

II. Standard Addition

An unknown sample of Ni^{2+} gave a current of 2.36μA in an electrochemical analysis (assume current is linearly proportional to concentration of Ni^{2+}). When 0.5 mL of a solution containing 0.0287 M Ni^{2+} was added to 25.0 mL of unknown, the current increased to 3.79μA.

(12 pts) a) Derive the expression:

$$\frac{[Ni^{2+}]_i}{[S]_f + [Ni^{2+}]_f} = \frac{\mu A_{Ni^{2+}}}{\mu A_{S+Ni^{2+}}}$$

Where $[Ni^{2+}]_i$ is the concentration of the unknown, and $[Ni^{2+}]_f$ is the concentration of the unknown in the sample diluted by the addition of standard and $[S]_f$ is the

concentration of the standard after adding it to the sample. μA_{Ni}^{2+} is the current resulting from the unknown Ni sample and μA_{S+Ni}^{2+} is the current measured after addition of the standard to the unknown.

(12 pts) b) Denoting the initial, unknown concentration as $[Ni^{2+}]_i$, write an expression for the final concentration $[Ni^{2+}]_f$ after 25.0 mL of unknown was mixed with 0.5 mL of standard. (This is sometimes called a dilution factor)

(12 pts) c) In a similar manner, denoting the initial, standard concentration as $[S]_i$, write an expression for the final concentration of the standard $[S]_f$ after 25.0 mL of unknown was mixed with 0.5 mL of standard.

(12 pts) d) Using the equation in part a) find the unknown concentration of nickel, $[Ni^{2+}]$ in the sample.

III Internal Standard

With the internal standard technique strictly ratios of concentrations and signals are taken so that differences due to different sensitivities between two different analytes and amounts of samples are divided out. Ratio equations result:

$$\left(\frac{[X]}{[S]}\right)_{unknown} = \left(\frac{S_X}{S_S}\right)_{unknown} \quad and \quad \left(\frac{[X]}{[S]}\right)_{standard} = \left(\frac{S_X}{S_S}\right)_{standard}$$

In these equations [X] is the concentration of element X, [S] is the concentration some other element, S, used as a standard. Signals of each are listed for each as S subscript. Two experiments are done, one with standard X and S giving the equation on the right, and another with known S and unknown X giving the equation on the left.

A solution was prepared by mixing 5.00 mL of unknown (element X) with 2.00 mL of a solution containing 4.13 ppm of standard (element S) and diluting to 10.0 mL. The measured signal ratio in an atomic absorption experiment (two lines are used, one for X and one for S) was:

$$\left(\frac{S_X}{S_S}\right)_{unknown} = 0.808$$ Where Sx is the signal due to X and Ss is the signal due to S

In a separate experiment, it was found that for equal concentrations of X and S, the signal due to X was 1.31 times more intense than the signal due to S.

(12 pts) Find the concentration of X in the unknown. (Don't forget to correct for dilution, when presenting your final answer.)